First-Stage and Second Sampling Design Selection of Census Tracts (PSUs)

The design follows the composite measure of size (MOS) framework for three-stage sampling described in Valliant, Dever, and Kreuter (2018, Sections 10.5.1 and Chapter 11). The notation is adapted from their More Than Two Stages of Sampling presentation (pp. 285–286).

Symbol	Definition
$d = 1, 2, 3, 4$	Age domains: 18–34, 35–49, 50–64, 65+
n_d	Desired number of respondents in domain d (= 250 for all d)
r_d	Expected response rate: 0.35, 0.45, 0.50, 0.65
$m_d = n_d / r_d$	Number of persons to be selected (inflated for nonresponse) in domain d
$Q(d)$	Total population of domain d in the Detroit frame (from Census data)
$Q_i(d)$	Population of domain d in tract i (summed over all BGs in tract i)
$f_d = m_d / Q(d)$	Desired overall sampling rate for domain d
$m = 40$	Number of sample PSUs (tracts)
$\bar{n} = 1$	Number of sample SSUs (BGs) per PSU
$\bar{\bar{q}}$	Expected workload (persons to be selected) per sample SSU

The number to be selected per domain, inflated for nonresponse:

Domain 1 (18–34)	Domain 2 (35–49)	Domain 3 (50–64)	Domain 4 (65+)
$m_1 = 250/0.35 \approx 714.29$	$m_2 = 250/0.45 \approx 555.56$	$m_3 = 250/0.50 = 500.00$	$m_4 = 250/0.65 \approx 384.62$

2. Composite Measure of Size (MOS)

To simultaneously achieve (1) a self-weighting sample within each age domain and (2) equal workload in each sample PSU, the composite MOS method is used (Valliant et al., 2018, Eq. 10.1, extended to three stages on pp. 285–286).

Tract-Level Composite MOS

The composite MOS for tract i is defined as:

$$S_i = \sum_d f_d \cdot Q_i(d)$$

(Valliant et al., Eq. 10.1)

This is the expected number of persons to be selected from tract i if the desired domain-specific rates f_d were applied directly.

Total Composite MOS

The total composite MOS across all tracts is:

$$S = \sum_{i \in U} S_i = \sum_d f_d \cdot Q(d) = \sum_d m_d$$

Key property: S equals the total number of persons to be selected across all domains (≈ 2,154.5).

Expected Workload per SSU

Since $S = m \cdot \bar{n} \cdot \bar{\bar{q}}$ (Valliant et al., p. 286), the expected workload per SSU is:

$$\bar{\bar{q}} = S / (m \cdot \bar{n}) = S / (40 \times 1) = S / 40$$

3. First-Stage Selection Probability

Tracts are selected with probability proportional to the composite MOS S_i:

$$\pi_{1i} = m \cdot S_i / S = 40 \cdot S_i / S$$

(Valliant et al., p. 286)

This ensures that when combined with the second- and third-stage probabilities, the overall selection probability for any person in domain d equals f_d, and the workload in each PSU is constant at $\bar{n} \cdot \bar{\bar{q}} = \bar{\bar{q}}$ (since $\bar{n} = 1$).

4. Method of Selection

Algorithm: Systematic PPS sampling, implemented via UPsystematic() in the R sampling package.
Sorting: The tract frame is sorted by tract GEOID (FIPS code) prior to selection, as required by the project specification. Sorting by geography before systematic PPS provides an implicit geographic stratification effect.
Random seed: set.seed(-77) is called before selection.
Number selected: m = 40 tracts.
Sampling interval: On the cumulative MOS scale, the interval is S / 40.

5. Frame Preparation and Quality Control

Before selection, the tract frame must be checked (Valliant et al., pp. 286, 308–309):
(a) Tracts with fewer than 2 BGs. If a tract contains only 1 BG, then the second-stage selection is deterministic ($\pi_{j|i} = 1$). This is not problematic per se, but such tracts should be flagged and potentially combined with an adjacent tract if they also have a small population.
(b) Tracts with small populations. Check whether $\pi_{1i} \leq 1$ for all tracts. If any tract has $\pi_{1i} > 1$, it must be treated as a certainty PSU or combined with a neighboring tract. Additionally, verify that $\bar{\bar{q}} \leq Q_i$ for every PSU (the total population of each tract must support the workload; Valliant et al., p. 286, check 3). Also ensure π_{1i} is not extremely small, which would lead to extremely large weights.
(c) Combining small units. Small tracts are combined using an operational adjacency rule based on the ordered tract identifiers, where each flagged tract is paired with the nearest available tract in the sorted list. This serves as a proxy for geographic adjacency in the absence of explicit spatial information. After combining, the composite MOS of the combined unit is defined as the sum of the individual MOSs (Kish linking procedure; Valliant et al., p. 286).

6. Summary of First-Stage Design Elements

Element	Specification
Sampling unit	Census tract (PSU)
Frame	All census tracts in the City of Detroit, after combining small tracts
Measure of size (MOS)	Composite MOS: $S_i = \sum_d f_d \cdot Q_i(d)$
Selection probability	$\pi_{1i} = 40 \cdot S_i / S$
Selection method	Systematic PPS (UPsystematic in R sampling package)
Sorting before selection	By tract GEOID (FIPS code)
Number of PSUs selected	m = 40
Sampling interval	On the cumulative MOS scale: S / 40
Quality checks	$\pi_{1i} \leq 1$; $\bar{\bar{q}} \leq Q_i$; no excessively small MOS

7. Verification of Design Properties

The design guarantees the following properties when both stages use pp(S) selection (Valliant et al., p. 286):
Requirement 1: Self weighting
The overall selection probability for any person k in domain d is:

$$\pi_{1i} \cdot \pi_{2j|i} \cdot \pi_{k|ij}(d) = (m \cdot \bar{n} \cdot S_{ij} / S) \cdot (\bar{\bar{q}} \cdot f_d / S_{ij}) = (m \cdot \bar{n} \cdot \bar{\bar{q}} / S) \cdot f_d = f_d$$

Requirement 2: Equal Workload per SSU
The expected total selections in any sample SSU is:

$$\sum_d q^*_{ij}(d) = (\bar{\bar{q}} / S_{ij}) \cdot \sum_d f_d \, Q_{ij}(d) = \bar{\bar{q}}$$

Requirement 3: Equal Workload per PSU

Since $\bar{n} = 1$, the workload per PSU is also $\bar{\bar{q}}$.

Second-Stage Sampling Design Selection of Block Groups (SSUs)

It follows the composite measure of size (MOS) framework for three-stage sampling described in Valliant, Dever, and Kreuter (2018, Section 10.5.1 and Chapter 11, pp. 285–286, 307–312).

1. Sampling Unit

The second-stage sampling unit (SSU) is the census block group (BG). Within each of the 40 sampled PSUs, exactly one BG is selected ($\bar{n} = 1$).

2. Frame

Data source. The BG-level data comes from the "BlockGroup" sheet of Detroit MI.xlsx (2020 Decennial Census DHC Summary File). This sheet has the same 52 columns as the "Tract" sheet—housing counts, total person counts, and person counts by sex × age—but recorded at the block group level. After removing duplicate records, there are 627 unique BGs across 276 tracts.

Identifying which tract a BG belongs to. Census GEOIDs follow a hierarchical coding structure: a BG GEOID is 12 digits composed of State (2 digits) + County (3 digits) + Tract (6 digits) + Block Group (1 digit). Therefore, the first 11 digits of any BG GEOID are exactly the GEOID of its parent tract. This is how BGs are linked to their parent tracts in the frame.

Constructing the sub-frame within each sampled PSU. In the first stage, some tracts were combined into composite PSUs (e.g., a tract with only 1 BG was paired with a neighboring tract). Each sampled PSU therefore consists of one or more original tracts. The second-stage frame for a given PSU is simply all BGs whose parent tract is one of the component tracts of that PSU. For a single-tract PSU, this is all BGs in that tract; for a combined PSU, it includes BGs from both tracts.

Every sampled PSU is guaranteed to contain at least 2 BGs, because the first-stage combining rule required every PSU to have at least 2 BGs. This ensures that the second-stage selection is non-deterministic in every PSU.

3. BG-Level Age-Domain Counts

The four age-domain population counts are constructed at the BG level using exactly the same variable definitions as in the first stage: Q_ij(1) for ages 18–34, Q_ij(2) for ages 35–49, Q_ij(3) for ages 50–64, and Q_ij(4) for ages 65+. Each is the sum of the corresponding male and female age-bracket columns from the P12 table. The total adult population in BG ij is TotAdult18plus = Q_ij(1) + Q_ij(2) + Q_ij(3) + Q_ij(4).

4. Composite MOS at the BG Level

Using the same domain-specific sampling rates f_d computed in the first stage (Valliant et al., Eq. 10.1, three-stage extension pp. 285–286), the composite MOS for BG j in PSU i is:

$$S_{ij} = \Sigma_d f_d \cdot Q_{ij}(d) = f_1 \cdot Q_{ij}(1) + f_2 \cdot Q_{ij}(2) + f_3 \cdot Q_{ij}(3) + f_4 \cdot Q_{ij}(4)$$

where the rates are:

Rate	Value	Derivation (m_d / Q(d))
f_1	0.004427	714.29 / 161,357

f_2	0.004905	555.56 / 113,272
f_3	0.004246	500.00 / 117,763
f_4	0.004384	384.62 / 87,739

Consistency property. The sum of BG-level MOSs within any PSU equals the PSU-level MOS that was used in the first stage: $S_i = \Sigma_j S_{ij}$. This has been verified with actual data (maximum discrepancy = 0). This property is what makes the two-stage composite MOS framework internally consistent—the first stage and the second stage are using the same underlying measure, just at different levels of aggregation.

Global constants carried from the first stage: S = 2,154.46 (total composite MOS = total number of persons to be selected across all domains); $\bar{\bar{q}} = S / (m \cdot \bar{n}) = 2{,}154.46 / (40 \times 1) = 53.86$ (expected workload per sample SSU).

5. Quality Control Within Each Sampled PSU

Before performing the second-stage selection, each BG in the PSU's sub-frame must pass the quality control checks specified in Valliant et al. (p. 286, checks 1–4):

(a) Workload feasibility: $\bar{\bar{q}} \leq$ TotAdult18plus for each BG. The expected workload per SSU ($\bar{\bar{q}} \approx 53.86$) must not exceed the total adult population of the BG. In the full Detroit BG frame, 29 BGs violate this condition (including 15 BGs with zero adult population, i.e., $S_{ij} = 0$). If any such BGs fall within a sampled PSU, they must be combined with a neighboring BG in the same PSU (identified by GEOID adjacency). After combining, the merged unit's MOS is the sum of the component MOSs.

(b) Domain-level feasibility: $q^*_{ij}(d) \leq Q_{ij}(d)$ for every BG and domain. The expected number of selections in domain d must not exceed the population count in that domain. This is equivalent to checking $f_d \leq S_{ij} / \bar{\bar{q}}$.

(c) Selection probability check: $\pi_{2j|i} \leq 1$. This is automatically satisfied as long as the PSU contains at least 2 BGs with positive MOS.

(d) Zero-MOS BGs must be removed or combined. A BG with $S_{ij} = 0$ has no adults in any age domain and cannot support any workload.

6. Conditional Selection Probability

Within sampled PSU i, BG j is selected with probability proportional to its composite MOS (Valliant et al., p. 286). Since $\bar{n} = 1$:

$$\pi_{2j|i} = \bar{n} \cdot S_{ij} / S_i = S_{ij} / S_i$$

where S_i is the PSU-level composite MOS (already computed and recorded in the first stage). The conditional probability is simply each BG's share of the PSU's total MOS.

7. Selection Method, Sorting, and Random Seed

Method. Systematic PPS sampling, using UPsystematic() from the R sampling package. This is the same method used in the first stage and is explicitly required by the project specification: "Use systematic sampling with probabilities proportional to size to select the PSUs and SSUs."

Sorting. Within each sampled PSU, BGs are sorted by their 12-digit GEOID before applying systematic PPS. This mirrors the first-stage requirement ("Sort the file in order by tract before selecting the sample") and provides an implicit geographic ordering within each PSU.

Random seed. set.seed(-77) is called once at the beginning of the entire sampling script, before the first-stage selection. The random number generator state then flows sequentially from the first-stage UPsystematic() call into the 40 second-stage calls. The seed must not be reset between stages; doing so would break the reproducibility chain.

Procedure. For each of the 40 sampled PSUs, the second-stage selection proceeds as follows: (1) identify all BGs belonging to the PSU's component tracts; (2) compute each BG's composite MOS and conditional selection probability; (3) perform quality control checks and combine deficient BGs if necessary; (4) sort remaining BGs by GEOID; (5) apply systematic PPS to select 1 BG.

8. Number Selected

1 BG per PSU × 40 PSUs = 40 BGs total.

9. Joint Selection Probability and Base Weight

The joint (unconditional) selection probability of BG j in PSU i, across both stages:

$$\pi_{ij} = \pi_{1i} \cdot \pi_{2j|i} = (40 \cdot S_i / S) \cdot (S_{ij} / S_i) = 40 \cdot S_{ij} / S$$

The PSU-level MOS S_i cancels in the product. As Valliant et al. (Chapter 11, p. 308) observe, this is algebraically equivalent to selecting 40 BGs directly from the full BG frame with probability proportional to S_{ij}. However, the two-stage clustered design differs in its joint inclusion probabilities: two BGs in the same PSU can never both appear in the sample, whereas a direct PPS draw of 40 BGs would give all pairs nonzero joint selection probabilities.

Base weight: $w_{ij} = 1 / \pi_{ij} = S / (40 \cdot S_{ij})$.

10. Derived Person Sampling Rates

The project appendix requires that for each sample SSU, "the sampling rate you will use to select persons in each domain" be listed. These rates are not an independent design decision—they are fully determined by the composite MOS framework once the first and second stages are complete. Specifically, once a BG is selected, its composite MOS S_{ij} is known, and the within-BG sampling rate for persons in domain d is (Valliant et al., p. 286):

$$\pi_{k|ij}(d) = \bar{\bar{q}} \cdot f_d / S_{ij}$$

The expected number of persons to be selected from domain d in BG ij is $q^*_{ij}(d) = \pi_{k|ij}(d) \cdot Q_{ij}(d)$. These expected counts are not integers. Persons are sampled at the fixed domain-specific rates above, not at fixed sample sizes (Valliant et al., Chapter 11, p. 310).

These four rates (one per age domain) will be reported for each of the 40 selected BGs in the SAMPLE output file.

11. Design Property Verification

Self-weighting. The overall selection probability for any person k in domain d, combining all three stages, is (Valliant et al., p. 286):

$$\pi_{1i} \cdot \pi_{2j|i} \cdot \pi_{k|ij}(d) = (40 \cdot S_i / S) \cdot (S_{ij} / S_i) \cdot (\bar{\bar{q}} \cdot f_d / S_{ij}) = 40 \cdot \bar{\bar{q}} \cdot f_d / S = f_d$$

The last step uses $S = 40 \cdot \bar{q}$. Every person in domain d has the same overall selection probability f_d, regardless of which tract or BG they reside in.

Equal workload. The expected total number of persons to be selected in any sample BG is:

$$\Sigma_d\, q^*_{ij}(d) = (\bar{q} / S_{ij}) \cdot \Sigma_d\, f_d \cdot Q_{ij}(d) = (\bar{q} / S_{ij}) \cdot S_{ij} = \bar{q} \approx 53.86$$

The workload is constant across all sample BGs and, since $\bar{n} = 1$, also across all sample PSUs.

12. Summary of Second-Stage Design Elements

Element	Specification
Sampling unit	Census block group (SSU)
Frame	All BGs within each sampled PSU's component tracts
Measure of size	Composite MOS: $S_{ij} = \Sigma_d f_d \cdot Q_{ij}(d)$
Conditional selection probability	$\pi_{2j\|i} = S_{ij} / S_i$
Joint selection probability	$\pi_{ij} = 40 \cdot S_{ij} / S$
Selection method	Systematic PPS (UPsystematic in R)
Sorting before selection	By BG GEOID within each PSU
Number selected	1 BG per PSU, 40 BGs total
Quality checks	$\bar{q} \le Q_{ij}$; $q^*_{ij}(d) \le Q_{ij}(d)$; all $\pi \le 1$
Derived person sampling rate	$\pi_{k\|ij}(d) = \bar{q} \cdot f_d / S_{ij}$

13. Output Files (per Project Deliverables)

FRAME2 file: All BGs within the 40 sampled PSUs, with each BG's GEOID, parent tract GEOID, parent PSU identifier, census population counts, age-domain populations, composite MOS (S_{ij}), and conditional selection probability ($\pi_{2j|i}$).

SAMPLE file: The 40 selected BGs, with all the FRAME2 variables plus: first-stage selection probability (π_{1i}), joint selection probability (π_{ij}), base weight ($w_{ij} = 1/\pi_{ij}$), and the four domain-specific person sampling rates $\pi_{k|ij}(d)$ for d = 1, 2, 3, 4.